

AUTHOR

Agentic Understanding for Thesis, Hypothesis, and Objective Research

One host-invocable package across the whole scientific-paper lifecycle:

read -> understand -> verify citations -> review -> plan -> write -> illustrate -> export

Invoked by the CLIO agent as a subagent . 26 actions . grounded, not hallucinated

1 . What this is (and the paper)

- **AUTHOR:** one Python package that turns a paper (arXiv link / PDF / title) into clean Markdown + memory blocks, then runs specialized expert agents to answer, verify, review, plan, write, illustrate, and export - behind one interface a host agent can call.
- **Paper title:** AUTHOR - Agentic Understanding for Thesis, Hypothesis, and Objective Research.
- **Core claim:** the contribution is unifying the lifecycle in one grounded, composable package - not any single capability in isolation.
- **Status:** built end-to-end - 26 actions, ~579 hermetic tests, CI green; validated live (Claude / Codex / Ollama / Gemini vision / live citation backends).

2 . Motivation - why this is needed

- Scientific output is exploding while the tools to process it stay siloed - reading, checking, and writing papers is still mostly manual or glued together by hand.
- **Fragmentation tax:** a researcher stitches together a PDF parser, a literature-QA tool, a citation auditor, a writing agent, a figure agent, and a LaTeX exporter - none sharing a data model, each re-parsing the same paper.
- **Hallucination is the central failure mode:** LLM writing fabricates citations at high rates (78-90% reported for GPT-4o); survey generators are uniformly weak on reference quality. Adoption demands grounding, not free-form generation.
- **Closed pipelines don't compose:** end-to-end 'AI scientist' systems emit one artifact and stop - a host agent can't reach in and invoke just the review step, or just ingestion.
- **Hosts need an on-demand capability:** a larger agent (CLIO) should load paper processing/writing, use it, and drop it - like calling a library, not running a monolith.
- **One substrate, reused:** ingest once -> the same memory blocks feed Q&A, review, planning, and writing - no lossy re-parsing between stages.

2b . The cost today (concretely)

- A typical 'process + review + write related work' task today touches 4-6 disjoint tools:
 - . PDF -> Markdown: Docling / MinerU (no writing, no citation check)
 - . Ask questions about it: PaperQA2 / OpenScholar (separate index, separate API)
 - . Check the references: a citation auditor (CiteCheck) - yet another tool, another format
 - . Draft text: a survey/writing agent (AutoSurvey / PaperOrchestra) - starts over, re-ingests
 - . Make a figure: PaperBanana; export LaTeX: a sixth step
- Each boundary loses structure, repeats work, and adds a place for hallucinated or unverifiable output to slip in. Nothing carries the paper's grounded representation across all stages.
- AUTHOR replaces that chain with one grounded substrate and one interface - the integration is the point.

3 . The gap - nobody unifies the lifecycle

Camp	Examples	Does well	Omits
Ingest-only parsers	Docling, MinerU	PDF->MD + vision	write / review / verify / host API
End-to-end generators	AI Scientist, Agent Lab, CycleResearcher	ideate->experiment->write (closed)	external ingest, cite-verify, host API
Single-slice specialists	AutoSurvey, PaperBanana, CiteCheck, PaperC	one capability, very well	every other capability

The quadrant {vision ingest + grounded QA + cite-verify + review + write + figures + export} as ONE host-invokable package is held by no one. That is AUTHOR's position.

3b . Gap - even the closest systems miss most of it

- AI Scientist v2 - generates NEW research end-to-end and reviews its own output, but does not ingest external PDFs, does no grounded Q&A, does not verify citations against scholarly DBs, and is a closed CLI with no host tool-surface.
- PaperOrchestra - strongest writing system (write + figures + export), but: no ingest, only partial QA, only partial citation-verify, no review, no host API.
- AutoSurvey / SurveyForge - write cited surveys from a curated abstract DB; cannot ingest arbitrary PDFs with vision, do not verify against external DBs, no review/figures/export.
- PaperQA2 / OpenScholar - excellent grounded literature-QA and host-invocable, but verify their OWN answers against a corpus, not a manuscript's bibliography; no ingest-with-vision, review, writing, figures, or export.
- **Docling / MinerU - best-in-class ingest, then stop:** nothing downstream.
- **Takeaway:** the capabilities exist but are disjoint across systems and don't interoperate - exactly the integration gap AUTHOR closes.

4 . Capability coverage (AUTHOR vs the field)

System	In	QA	CV	Rv	Wr	Fg	Ex	1pkg	Host
AUTHOR	Y	Y	Y	Y	Y	Y	Y	Y	Y
AI Scientist v2	N	N	P	Y	Y	Y	Y	P	N
PaperOrchestra	N	P	P	P	Y	Y	Y	P	N
AutoSurvey	N	Y	P	N	Y	N	N	P	N
PaperBanana	N	P	N	P	N	Y	P	N	P
Docling / MinerU	Y	N	N	N	N	N	P	Y	Y
PaperQA2	N	Y	P	N	N	N	N	N	Y

No row but AUTHOR is all-Y. In=ingest QA=grounded-QA CV=cite-verify Rv=review Wr=write Fg=figures Ex=export. Full matrix + a source per cell: docs/MOTIVATION.md

5 . What we built - the 26 actions

- **Read / understand:** ingest (PDF->MD + blocks + figures), ask (grounded Q&A), kg (content knowledge graph)
- **Discover/verify:** discover (find real papers via scholarly search), cite (verify; 4-source cascade), check_refs (BibTeX audit)
- **Review:** review (decision + scores + --ground; multimodal - sees figures via vision), meta_review (panel), rebuttal (point-by-point), write_review (loop)
- **Write:** plan, write, edit, polish, coherence, compose (whole paper), export (-> LaTeX + PDF via --pdf)
- **Illustrate:** plot, describe_figures (Gemini vision), figure_refine (loop)
- **Research/verify:** research (grounded lit brief), verify_work (claim coverage), check_refs (BibTeX audit), section_review (3-layer), audit (pre-submission)
- **Knowledge graph:** kg (single-shot) and kg --full (6-stage pipeline: ontology->extraction->coref->verification, clean-room protoneo)
- **Drive:** orchestrate (goal -> plan a sequence of the above -> execute)
- One adapter (capabilities()) + run(action, payload)) + a CLI - stateless, JSON in/out, never raises.

6 . References & artifacts we built on

- Combined (the mandate):

- . Processing - paper-to-md (+ phagocyte) and PaperBanana (arXiv 2601.23265)
- . Writing / editing - wtf-p and PaperOrchestra (arXiv 2604.05018)

- Harness references (concepts re-implemented, not copied):

- . papervizagent - orchestrator + expert agents + critic loop -> CriticRefine + figure_agent
- . protoneo/knowledge (AGPL) - BaseAgent / deliberation patterns / KG -> harness/ + kg
- All cloned in artifact/repos/; both papers in artifact/papers/; deep-study notes in artifact/notes/.
- License-clean (BSD-3-Clause; no AGPL code copied - protoneo concepts re-implemented from scratch).

7 . Design - how it's put together

```
host (CLIO) ---> ClioAuthorSubagent (capabilities + run; JSON; never-raises)
                  |
                  Main agent (router) + orchestrate (goal -> plan -> execute)
                  |
expert agents:  ingestor . paper_qa . citation . reviewer/meta .
planner . writer/editor . polish . coherence . figure_agent . kg .
compose . export
                  |
backends:  retrieval (RAG + scholarly cascade) . vision (Gemini) .
SafeFiles (read/write/edit, sandboxed) . LLM (claude/codex/ollama/echo)
```

Principles: expert agents + retrieval + read/write/edit tools . grounded (verify, source-bound) . standalone harness, thin host bridge . hermetic-first (offline echo model; heavy paths gated).

8 . How we'll prove it - evaluation plan

Track	Why we need it	Baselines
PDF->MD fidelity	ingest errors propagate to every action	paper-to-md, Docling, MinerU
Figures	high-bar, easily-judged; backs 'writes a paper'	PaperBanana, DeTikZify
Citation verify	the grounding claim; hallucination is the top failure	CiteCheck, PaperOrchestra
Review	must track ground-truth decisions; does --ground help?	AgentReview, DeepReview (data: ASAP-Review, PeerRead, ORB, MMReview)
Writing	most visible output; parity = credible, not marketing	PaperOrchestra, AutoSurvey
Orchestration + co	composability is part of the novelty; budgets	-

Review datasets secured: PeerRead, ASAP-Review, NLPeer, MOPRD, ORB, MMReview (cross-domain + multimodal). Phasing: Phase 0 automated metrics (now) -> Phase 1 LLM/VLM-as-Judge -> Phase 2 human win-rates. Full plan: docs/BENCHMARK-PLAN.md

9 . Justification - why this is worth doing

- **A real, unoccupied gap:** surveyed 20+ systems; none unify the lifecycle as one grounded, host-invocable package (slides 3-4).
- **Fixes the trust problem:** verification + source-grounded writing vs hallucination-prone generators.
- **A reusable substrate, not a one-off:** a host (CLIO) gains paper read/review/write as composable, on-demand capabilities; experts compose in any order.
- **Already real:** built, tested, demonstrable today; the benchmark plan turns the claim into numbers a reviewer will accept.
- **External validation:** the newest cross-domain + multimodal review benchmark (MMReview) tests exactly our edge - vision-grounded review across 17 domains, which text-only reviewers structurally can't do.
- **Objection - 'isn't this just tool orchestration?':** frameworks (MCP, LangGraph) are domain-agnostic plumbing that ship zero paper-lifecycle capability. AUTHOR is the missing domain package; MCP is just how a host reaches it.

10 . Status & next steps

- **Done:** 26 actions . ~579 hermetic tests + CI . live-validated (Claude/Codex/Ollama/Gemini, real citation backends) . motivation + benchmark plan . dynamic orchestrate.
- **Next:**
 - . Demo - prove a host agent (Claude/Codex) invokes AUTHOR as a subagent end-to-end.
 - . CLIO integration - thin MCP bridge (CLIO invokes external capability via MCP); file a CLIO feature request for an in-process subagent hook.
 - . Benchmarks - run Phase 0 metrics; scope datasets / judge models / human eval.
- **Open decisions:** the bar (beat specialists vs parity + unification) . dataset access (PaperBananaBench / PaperWritingBench) . human-eval resourcing.
- **Repo:** github.com/SIslamMun/clio-author . docs/MOTIVATION.md, BENCHMARK-PLAN.md, RUNBOOK.md